StaffSync Project Blueprint

FULL-STACK MERN HUMAN RESOURCE MANAGEMENT SYSTEM

Developer Roadmap

To complete your StaffSync project effectively on the platform, follow this structured roadmap aligned with the project's milestones, epics, and engineering requirements.

STEP 1 UNDERSTAND PROJECT SCOPE & ARCHITECTURE

Core Goal: Build a complete MERN stack (MongoDB, Express, React, Node.js) Human Resource Management (HRM) system delivering role-governed administrative workflows across 6 Epics and 19 Tasks.

HR Administrator

Manages complete employee records, reviews leave requests, monitors daily attendance logs, and publishes organization announcements.

Team Manager

Supervises reporting employees, approves/rejects team leave requests, and coordinates departmental training and skill development.

Employee

Applies for planned leaves, marks daily workplace attendance, tracks approval progress, and reads company bulletin announcements.

Epic	Phase & Module Focus	Key Deliverables
01	Project Architecture & Setup	Monorepo/dual-repo structure, boilerplate, package setup, tooling
02	Backend Development	Node.js & Express RESTful APIs, JWT role-based authentication
03	Database Development	MongoDB schemas for Users, Attendance, Leaves, Announcements
04	Frontend Development	React functional components, React Router, responsive Bootstrap UI
05 & 06	Execution, Testing & Deployment	API validation, cloud deployments (Render, Vercel), mentor review

STEP 2 SET UP YOUR LOCAL DEVELOPMENT ENVIRONMENT

Node.js (v16 or above)

Runtime environment for executing backend services and powering Vite/React frontend build scripts.

MongoDB Community / Atlas

Local database server instance or cloud-managed cluster configured with secure IP whitelist and connection strings.

Visual Studio Code

Primary IDE configured with ESLint, Prettier, and standard MERN development extensions.

Postman API Client

Tool for testing authentication tokens, validating request payloads, and executing endpoint collections.

STEP 3 FOLLOW THE DEVELOPMENT WORKFLOW

1. Project Setup & Configuration

Foundation

- **Backend:** Initialize server repository (`npm init -y`), configure core packages (`express`, `mongoose`, `cors`, `dotenv`), and define clear environment variables.
- **Frontend:** Initialize client application (`create-react-app` or Vite), configure React Router, and install UI styling libraries such as Bootstrap.

2. Database Design (MongoDB & Mongoose)

Data Layer

- **User Schema:** Fields for `name`, unique `email`, `password` (securely hashed via `bcrypt`), and `role` (HR, `Manager`, `Employee`).
- **Leave Schema:** Fields for `employeeId` (ObjectId ref), `startDate`, endDate, `reason`, and `status` (`Pending`, `Approved`, `Rejected`).
- **Attendance & Announcement Schemas:** Structured collections to record timestamps, daily presence states, and org-wide broadcast notices.

3. Backend Development (RESTful API Endpoints)

API Layer

- **Authentication:** Build `/api/auth/register` and `/api/auth/login` incorporating signed JSON Web Tokens (JWT) and role verification middleware.
- **Business Logic CRUD:** Expose protected endpoints to manage staff listings, post company announcements, and submit/review leave requests.

4. Frontend Development (React Application UI)

Interface

- **Role Dashboards:** Build dedicated portal views customized with contextual capabilities for HR, Managers, and Employees.
- **Protected Routes:** Restrict client navigation guards so unauthorized roles cannot view administrative features.
- **Data Integration:** Bind UI components to Express endpoints via `fetch` or `axios` with bearer token state persistence.

STEP 4 SUBMIT YOUR WORK FOR EVALUATION

1. **Version Control:** Push the thoroughly tested codebase with clean commit history to a public or private GitHub repository.
2. **Cloud Deployment:** Host the client on Vercel or Netlify and the API service on Render or Railway with production environment variables set.
3. **Platform Submission:** Access your project dashboard, click `[Add Demo Link]` and `[Add Github Link]`, and submit your entries for formal mentor evaluation.

Ready to Begin Development?

Next recommended steps include generating the Mongoose data schemas or authoring the role-based JWT authentication middleware.